có nhiều cách viết - đây là một cách viết bạn tham khảo - bài viết ở dạng short
- bài viết dài cần triển khai và mở rộng nhiều ý dựa trên cách trình bày này.

cách đặt tên cho bài báo

áp dụng vào bài toán gì

Voice Separation using Multi-Learning on Squash-norm Embedding Matrix and Mask

phương pháp đề xuất hay dựa trên phương pháp áp dụng
(lưu ý phương pháp này tự đề xuất hay dựa vào phương pháp đã có)

No Author Given

No Institute Given phần tác giả của bài viết tại đây

Abstract. Recently, deep learning has achieved state-of-the-art performance for many fields e.g., image classification, action recognition, natural language processing, speech recognition, etc. For the speech separation issue, conventional networks directly optimize sources or masks while deep clustering learns embedding matrix training during and clusters an embedding matrix testing during. Deep clustering has obtained some outstanding results in time frequency domain speech separation. Conventional networks have an end-to-end training to take advantage directly signal approximation. The properties and strengths of the traditional networks and deep clustering are combined to improve performances. In this study, we have proposed a multi-learning method by combining a deep clustering and one conventional network. The experiment was conducted on the TIMIT dataset - one of the most standard datasets for speech separation and the results have shown that this method outperforms the component network and state-of-the-art methods recently proposed in terms of common metrics like SDR, SIR, and SAR.

phần màu vàng là giới thiệu tổng quát

Phần màu biển là phân tích các công nghệ nên tảng, điểm hay và hạn chế, để đưa ra phương pháp của chúng ta.

Phần màu xanh lá là đề xuất công nghệ để khắc phục các công nghệ trước đây

Phần màu đó là kết quả đạt được hiệu quả thế nào, phạm vi trên tập dữ liệu nào và có hơn phương pháp nào không?

Keywords: Speech separation, multi-learning, deep learning, embedding vectors, deep clustering, time frequency domain.

Phần Keywords là các từ khóa liên quan

1 Introduction

The recent development of deep learning models has revolutionized traditional machine learning applications with unprecedented performance gains coupled with the ability to self-training and represent complex information. Deep neural networks have been successfully applied in many fields including image classification [5, 13], object detection [4], action recognition [23, 2], video prediction [14, 24], etc. Deep learning has been constructed for voice separation, e.g., deep neural networks (DNNs), recurrent neural networks (RNNs) [7], [21] and convolutional neural networks (CNNs) [9]. In particular, deep clustering (DC) [6] has been introduced as an effective method for discrimination between the vocal sources. The Deep attractor network (DANet) [1] trains the attractor points in an embedding matrix. The permutation invariant training (PIT) [25] has been proposed to solve the permutation issue. Discriminative vector learning (DVL) method [18], [17] perform to train and cluster the embedding vector. Time-domain audio separation network (TasNet) [12] obtains a performance improvement. The models directly execute time-domain-based audio separation with the state-of-the-art

phần màu vàng viết dẫn nhập các thành công của công nghệ này, và các công trình quan trọng kèm theo trích dẫn

Phần màu xanh giới thiệu và phân tích các công nghệ nên tải để đề ra phương pháp đề xuất (tùy theo viết ngắn or dài, chúng ta phân tích sơ lượt hoặc kỹ ở phần này, tuy nhiên nó phải hướng vào công nghệ chúng ta đề xuất

performance, e.g., Wavesplit [26]. In addition, some lightweight models are proposed to suit low-resource mobile devices, e.g., Selective Mutual Learning [19], GroupComm-DPRNN [11], SeliNet [20], Sandglasset [10], etc.

DC enhances the performance and addresses the permutation invariant issue. However, the clustering process is still not really effective for the estimated mask. In this study, we proposed a multi-learning method on Squash-norm embedding feature matrix and mask. We design hybrid network to incorporate DC and the inference mask network. This hybrid multi-learning have own loss functions. DC trains the embedding matrix containing the embedding vectors with component sources' information. In particular, Squash-norm is adopted to normalize the embedding matrix to improve the discriminative ability of the deep vectors. The embedding elements are pulled closer together in the same partition while being pushed further away in two different partitions. We explore our designed model to increase the performance with respect to the component networks. The multi-learning method has two main contributions. First, Squash-norm are adopted to enhance the discrimination between the matrix's elements to increase the performance. Second, we train the embedding features through the multi-learning that improves the performance for the entire model.

In the rest of the study, we present a multi-learning on the Squash-norm matrix and mask in Section 2. Experiments are presented in Section 3. Finally, we summarize the work to do in the conclusion (in Section 4).

Câu màu xanh lá là dẫn nhập cho hạn chế để dẫn đến đưa ra công nghệ đề xuất.

Đoạn màu đỏ là đưa ra công nghệ thực hiện

Đoạn màu vàng là phân tích công nghệ của chúng ta đoạn này rất quan trọng, phải thật sự rõ.
(-trong bày viết ngắn này không thêm bình luận kết quả trong tập dữ liệu nào kết quả thế nào do bị giới hạn số trang - tuy nhiên viết dài phải triển khai và bình luận kết quả ở đoạn này)

Phân đoạn màu xanh này để người đọc tiện theo nắm bắt phần còn lại

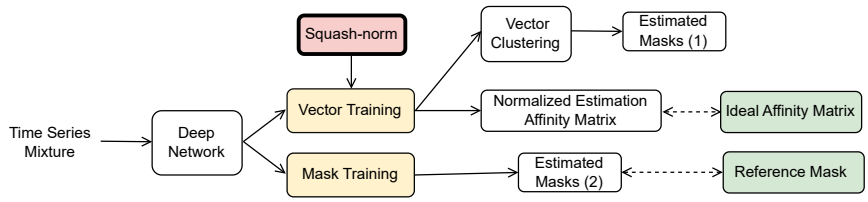


Fig. 1. Multi-learning method on Squash-norm embedding matrix and mask. Squash-norm is adopted to normalize the output-embedding vector. The multi-learning method is contributed by the loss function, which is the combination of the difference between the normalized estimation affinity matrix and ideal affinity matrix as well as the difference between the estimated mask and reference mask, respectively.

Hình 1: mô tả phương pháp và hình vẽ, sơ đồ khối, chỉ rõ ra phần công nghệ đề xuất

2 Multi learning on Squash-norm Matrix and Mask

Squash-norm Embedding Matrix: The vectors X_i are described as input features. Index i, $1 \leqslant i \leqslant N$ conforms to the time frequency (TF) index (t, f). The bidirectional long-short term memory (BLSTM) layers train the input mixture and the outputs are the deep embedding vectors with D dimension corresponding to TF bins. The vectors $v_i \in \mathbb{R}^{1\times D}$, $1 \leqslant d \leqslant D$, are stacked to construct the embedding matrix $V = [v_1, v_2, ..., v_N]$, $V \in \mathbb{R}^{N\times D}$. A unit-norm function is used

Phần phương pháp ĐỀ XUẤT có 3 ý chính:
+ MULTI LEARNING ;
+ SQUASH-NORM MATRIX;
+ VÀ MASK
DO VIẾT DẠNG NGẮN NÊN BỎ PHẦN MASK - VÌ PHẦN MASK ĐÃ CÓ TRƯỚC ĐÂY - NÊN 2 PHẦN QUAN TRỌNG LÀ:
+ MULTI LEARNING ;
+ SQUASH-NORM MATRIX;
PHẢI ĐƯỢC TRÌNH BÀY CẨN THẬN TRONG PHẦN PHƯƠNG PHÁP.

to normalize the embedding vector to DC [6]. In this paper, the mathematical function [15] is replaced to normalize the high-dimensional embedding vector v_i, and is computed as follows:

$$u_i = \frac{||v_i||^2}{1 + ||v_i||^2} \frac{v_i}{||v_i||}, \tag{1}$$

here $u_i \in \mathbb{R}^{1\times D}$ indicates the normalized embedding vector of normalized matrix $U \in \mathbb{R}^{N\times D}$. The embedding element $v_{i,d} \in \mathbb{R}^{1\times 1}$ is the point of the vector, v_i, with $||v_i||^2 = \sum_{d=1}^{D} v_{i,d}^2$ and $||v_i|| = \sqrt{||v_{i,d}||^2}$. Therefore, the normalized vector's length u_i ranges $[0; 1]$. The normalized embedding vector u_i dilates to an unit vector if the length of the embedding vector v_i is a large value. Conversely, u_i shrinks to close to the zero vector if the length of v_i is a small value. From equation (1), namely Squash-norm. This function ensures the discriminative learning between the normalized vectors. Equation (1) transforms V to $U = [u_1, u_2, ..., u_N]$. U is adopted in the clustering process, where the number N is the product of the time frame number T and the frequency bin number F. The permutation issue is handled in the training stage when the reference ratio affinity matrix $YY^T \in \mathbb{R}^{N\times N}$ is used to make the training target. The affinity matrix is converted to the ideal ratio mask $Y \in \mathbb{R}^{N\times C}$. The objective function estimates the difference between the reference ideal affinity matrix YY^T and the normalized estimation affinity matrix $UU^T \in \mathbb{R}^{N\times N}$ with C number of sources, is calculated as follows:

LƯU Ý TRONG PHƯƠNG PHÁP PHẢI PHÂN TÍCH CẨN THẬN VÌ NÓ LÀ NÊN TẢI CỦA CÔNG NGHỆ - ĐỐI HỎI PHẢI HIỂU PHƯƠNG PHÁP ĐỀ XUẤT VÀ TÍNH MỚI

$$\mathcal{L}_{DC} = \left\|UU^T - YY^T\right\|_F^2 \tag{2}$$

In order to reduce complex calculations by considering the low-rank issue of sparse element Y. This approximation infers to the objective function that we can calculate as follows:

$$\mathcal{L}_{DC} = \left\|U^T U\right\|_F^2 - 2\left\|U^T Y\right\|_F^2 + \left\|Y^T Y\right\|_F^2, \tag{3}$$

where $\|\,.\,\|_F^2$ denotes the squared Frobenius norm of the matrix. The normalized deep embedding vectors assigning to the single source signals in the mixture signal, which are clustered by the clustering algorithms in the testing stage. The estimated masks are constructed to recreate the estimated sources in the TF domain. Then, TF domain estimated sources are converted into a time series through the inverse Fourier transform.

Multi-learning Approach: Multi learning is built to combine the benefits of a mask network and DC. Using BLSTMs with two outputs, the first direct output from the last layer is a linear layer while another output is through U. We perform a loss with a masked magnitude spectrum approximation (MSA) for the output. This loss computes the permutation's cost and chooses the minimum. It is also referred to as permutation invariant training and is defined as

$$\mathcal{L}_{MSA} = \|(M - M_{MSA}) \odot X\|_2^2, \tag{4}$$

where $\| \cdot \|_2^2$ is the squared Frobenius norm, the $\odot$ operator (Hadamard product) indicates an element-wise multiplication. X presents the log-scale spectral magnitude of the single-channel mixture signal and M denotes the reference ideal ratio mask that is utilized to optimize the estimated mask of magnitude spectrum approximation M_{MSA}. We use Eq. (3) to learn knowledge from difference between the reference ideal affinity matrix and the normalized estimation affinity matrix. The total loss is defined as the weighted sum of the DC loss $\mathcal{L}_{DC}$, and the magnitude spectrum approximation loss $\mathcal{L}_{MSA}$, and is written as

$$\mathcal{L} = \alpha \mathcal{L}_{DC} + \beta \mathcal{L}_{MSA} \tag{5}$$

where α, and β are the weights with $\alpha + \beta = 1$, which control the loss functions.

3 Experiments

phân thực nghiệm phải đủ các bước sau

Configuration, Regularization, and Metrics: Four BLSTMs with 600 units per each layer adopt $D = 40$, and the source's number of $C = 2$. Adam function uses $\eta = 0.0001$, $\beta_1 = 0.9$, $\beta_2 = 0.999$ and $\varepsilon = 1e-8$. The updated learning rate reduces the value by 10% if the performance does not increase over 10 epochs during training. The input's dropout is set to 50% and the output's dropout is set to 20% for the regularizing model. The value metrics are the source to artifacts ratio (SAR), source to interference ratio (SIR), source to distortion ratio (SDR), scale-invariant signal-to-noise ratio improvement (SI-SNRi), short-time objective intelligibility (STOI), and perceptual evaluation of speech quality (PESQ). Higher metrics indicate quantitatively better speech separation.

Configuration, Regularization, and Metrics:
- do viết ngắn nên rút gọn các phần này lại,
+ nói rõ cấu hình thực nghiệm; rất quan trọng, để biết được thí nghiệm thiết lập thế nào
+ Phần chỉ số đo lường đánh giá kết quả cũng được trình bày trong phần này

Experimental Setup: The training, validation, and test datasets are female and male speeches mixed at 0 dB SNR. The mixture is set to 129 dimensions extracted for 256-point DFT, utilizing the log-scaled magnitude spectrum with a window size of 32 ms, sampling frequency $f_s = 8$ kHz, hop length of 8 ms, Hanning window, and 100 frame sequence training features.

phần thiết lập dữ liệu thí nghiệm quan trọng, để biết được đào tào tập train, tập test, tập val - viết dài hơn cho loại bài viết nhiều trang

Comparisons and Results: As shown in Table 1, our separation performance (i.e. $\alpha = \beta$) is higher other methods on the TIMIT dataset. We mainly present the T-F domain-based methods. The proposed approach learns bringing significant efficiency when knowledge is accumulated from many different objectives.

Phần so sánh kết quả rất quan trọng nó là chứng minh phạm vi của phương pháp được viết kỹ

bình luận kết quả bảng 1: bảng này bình luận phương pháp đề xuất tốt hơn thế nào so với các phương pháp khác trên cùng dữ liệu và thiết lập thí nghiệm

Table 1. Comparisons of the values (dB) of the different models on the TIMIT corpus.

Method	SDR (dB)	SIR (dB)	SAR (dB)
DF-DNN [3]	6.39	9.72	9.89
CNN [22]	6.60	9.50	10.20
DRNNs [16]	7.74	-	-
RNN [8]	7.41	11.76	7.54
Our Model	**11.76**	**19.12**	**12.57**

bảng 1 là chúng ta so sánh để thấy tốt hơn các phương pháp khác.
trong khi hình 2 (fig. 2) là phần quan trọng nhất của phương pháp, chứng minh kết quả tốt hơn một phương pháp nền (ví dụ; bạn chọn phương pháp a++ dựa trên phương pháp a, và nó tốt hơn phương pháp a)

In Fig 2, changing the α from 0 to 1, we observe the different effects of learning from the output embedding matrix (EM) and learning from the reference mask (RM). Through the results obtained learning from different instructions has resulted in more separation of the cases of independent training. We see that the model's efficiency increases as knowledge is learned from EM and RM.

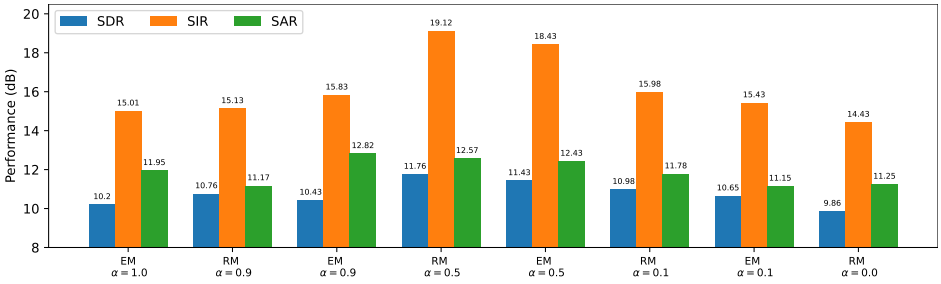


Fig. 2. The performances of voice separation. We investigate the influence of α and β coefficients on the model's loss function ($\alpha + \beta = 1$).

Bảng 2 (màu xanh) bình luận kết quả có squash-norm tốt hơn unit-norm - một chứng minh quan trọng trên phạm vi dữ liệu này

Table 2, using Squash-norm, the output feature elements are distinguished, and it is possible to pull the elements of the same group together and the elements of different groups away from each other. Experiments have also shown that this function improves the separation results in multi-task learning.

Table 2. Results for unit-norm and Squash-norm with $\alpha = \beta$ on the TIMIT corpus.

Normalization method	SDR	SIR	SAR	SI-SNRi	STOI	PESQ
Unit-norm	11.38	18.92	12.16	10.41	0.90	2.12
Squash-norm	**11.76**	**19.12**	**12.57**	**10.65**	**0.90**	**2.21**

hình 3 là kết quả bổ sung, và các thực nghiệm và hình phổ thấy được kết quả đạt được, bình luận.

In Fig. 3, the spectrums of the samples of the test dataset are shown. Fig. 3(a) shows the mixture of the female voice `{./FAKS0/SI1573.WAV}` and the male voice `{./MWBT0/SX383.WAV}`. Fig. 3(b) and (c) correspond to the spectral magnitudes of the female voice and the recovered female voice, respectively. Fig. 3(d) and (e) are the male voice's spectral magnitudes and the recovered male voice's spectral magnitudes, respectively. The male voice's recovered signals and female voice's recovered signals are learned by the designed model, with $\alpha = 0.5$, $\beta = 0.5$.

4 Conclusion

phần kết luận, tổng kết lại và đưa ra kết luận, hướng phát triển tương lai.

In this study, we have proposed multi-learning method on Squash-norm feature matrix and mask. We use the hybrid network to incorporate the DC and the infer-

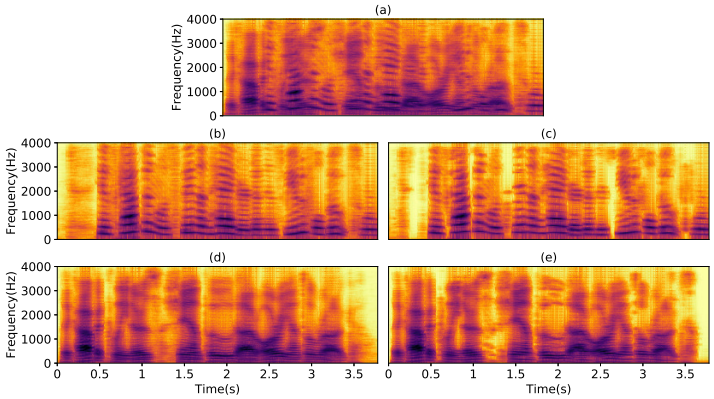


Fig. 3. The spectrograms of multi-learning method with $\alpha = 0.5$, and $\beta = 0.5$.

phần kết luận có 4 đoạn
thứ nhất: nói lại công nghệ đưa ra;
thứ 2: bình luận công nghệ hay thế nào
Thứ 3: Kết quả thực nghiệm và phạm vi
Thứ 4: hướng phát triển tương lai tiếp theo.

ence mask network with two main contributions. First, Squash-norm is adopted to improve the discriminative ability of the deep vectors. Second, we train the embedding features through the multi-learning that improves the performance for the entire model. Additionally, we have investigated performances with different clustering methods. Experiments have shown that the multi-learning method achieves higher performance than the component networks. In future works, we aim to further develop the designed architectures to enhance the ability for speech separation by proposing advanced methods and new objective functions.

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phân tài liệu tham khảo quan trọng, các công trình liên quan, và nền tảng, đọc và phân tích tài liệu rõ ở các mục.

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